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E-Learning Material

Fundamentals of Mathematics

(25VBT0C102)

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Module: 05

Three-Dimensional Geometry

Module Information

Field	Details
Course Title	Fundamentals of Mathematics
Course Code	25VBT0C102
Module Number	Module 5 of 5
Module Title	Three-Dimensional Geometry

Module Overview

Three-dimensional geometry is the branch of mathematics that extends the familiar ideas of two-dimensional coordinate geometry — points, lines, distances, and angles — into the full richness of three-dimensional space. Where a line in two dimensions is determined by just two parameters (slope and intercept), a line in three dimensions requires six coordinates (or their compact equivalents in vector form). Where two lines in a plane either intersect or are parallel, two lines in three-dimensional space have a third possibility — they may be skew: non-intersecting and non-parallel, passing each other in different planes. These richer geometric relationships make 3D geometry both more challenging and more powerful than its 2D counterpart.

The module begins with the fundamental concept of direction in three-dimensional space. The direction of a line is described by its direction cosines — the cosines of the angles the line makes with each of the three coordinate axes — and by the related direction ratios, which are any set of three numbers proportional to the direction cosines. These provide the indispensable language for describing orientation in space.

From direction, we move to lines. Every line in three-dimensional space can be described in two fundamental forms: the vector equation, which describes the line as the set of all points reached by starting at a fixed point and moving any distance in a fixed direction; and the Cartesian (symmetric) equation, which expresses the same information in terms of the coordinates x, y, z. We derive both forms from first principles and develop fluency in converting between them.

The angle between two lines in 3D space is determined by the dot product of their direction vectors — a clean and elegant formula. Lines in space may intersect, be parallel, or be skew. For skew lines, a key geometric quantity is the shortest distance between them — the length of the unique common perpendicular. We derive and apply formulas for shortest distance in both vector and Cartesian form.

Planes are the natural counterpart of lines in three dimensions. Every plane is determined by three non-collinear points, or equivalently by one point and the normal direction perpendicular to the plane. We derive vector and Cartesian equations of planes under various conditions: given a point and normal vector, given three points, and given intercepts on the coordinate axes. The angle between two planes is the angle between their normal vectors, giving a formula parallel to that for lines.

The final topics integrate lines and planes: coplanar lines (when do two lines lie in a common plane, and what is that plane?) and the intersection of a line with a plane (finding the unique point where a non-parallel line meets a given plane). These topics require synthesising all the earlier concepts and applying them to solve geometric problems in space.

Throughout this module, a dual treatment is maintained: every concept is developed in both vector form (using position vectors, direction vectors, and the dot and cross products from Module 2) and Cartesian form (using explicit coordinate equations). This dual approach equips you to work fluently in whichever representation best suits a given problem — a hallmark of mathematical maturity at the BTL 4–6 level.

Learning Objectives

By the end of this module, you will be able to:

- Define and compute direction cosines and direction ratios of a line in three-dimensional space, and use the relationship $l^2 + m^2 + n^2 = 1$ to verify and convert between them.
- Derive and apply the vector and Cartesian equations of a straight line in 3D space, given a point and direction vector, or given two points on the line.
- Evaluate the angle between two lines using direction vectors, and determine when lines are parallel, perpendicular, or skew; compute the shortest distance between skew lines.
- Derive and apply vector and Cartesian equations of a plane in multiple forms — point-normal, three-point, intercept, and general — and compute the angle between two planes.
- Analyse conditions for coplanarity of two lines and find the equation of the plane containing them; determine the intersection point of a line with a plane.

Learning Outcomes

After completing this module, you will be able to:

- Compute direction cosines and ratios for any given line and verify the identity $l^2 + m^2 + n^2 = 1$; find the direction cosines given direction ratios.
- Write the equation of a line in both vector form $r = a + \lambda b$ and symmetric Cartesian form, and convert between them.
- Find the angle between two lines, test for parallelism and perpendicularity, and calculate the shortest distance between two skew lines using both vector and Cartesian formulas.
- Determine the equation of a plane in various standard forms and compute the angle between two planes and the distance from a point to a plane.
- Test whether two lines are coplanar, find the plane of coplanarity if they are, and locate the intersection point of a line with a plane.

Table of Topics

Sub-Unit	Topic	Description
5.1	Direction Cosines and Direction Ratios	Angles α, β, γ with axes; $l=\cos\alpha$, $m=\cos\beta$, $n=\cos\gamma$; $l^2+m^2+n^2=1$; direction ratios a:b:c
5.2	Direction Cosines of a Line Through Two Points	Formula for d.c.s from coordinates of two points; distance between them
5.3	Equation of a Line — Vector Form	$r = a + \lambda b$; point a, direction b; geometric meaning; parameter λ
5.4	Equation of a Line — Cartesian Form	Symmetric form $(x-x_1)/l = (y-y_1)/m = (z-z_1)/n$; two-point form
5.5	Angle Between Two Lines	$\cos\theta = \lvert l_1 l_2 + m_1 m_2 + n_1 n_2 \rvert$; parallel (d.r.s proportional); perpendicular (dot product = 0)
5.6	Skew Lines and Shortest Distance — Vector Form	Condition for skew lines; SD = $\lvert (b_1 \times b_2)\cdot(a_2 - a_1) \rvert / \lvert b_1 \times b_2 \rvert$
5.7	Shortest Distance Between Skew Lines — Cartesian Form	Determinant form; condition for intersection
5.8	Equation of a Plane — Vector Form	$r\cdot\hat{n} = d$; point-normal form $r\cdot n = a\cdot n$
5.9	Equation of a Plane — Cartesian Forms	General: $ax+by+cz+d=0$; intercept form; passing through three points
5.10	Angle Between Two Planes	$\cos\theta = \lvert n_1 \cdot n_2 \rvert/(\lvert n_1 \rvert \lvert n_2 \rvert)$; parallel and perpendicular planes
5.11	Distance from a Point to a Plane	$d = \lvert ax_1 + by_1 + cz_1 + d \rvert / \sqrt{(a^2+b^2+c^2)}$; distance between parallel planes
5.12	Angle Between a Line and a Plane	$\sin\theta = \lvert b\cdot n \rvert / (\lvert b \rvert \lvert n \rvert)$; condition for line parallel to plane; line in plane
5.13	Coplanar Lines	Condition for two lines to be coplanar; equation of plane containing them
5.14	Intersection of a Line and a Plane	Substituting line into plane equation; foot of perpendicular from point to plane

5.1 Direction Cosines and Direction Ratios

In two-dimensional geometry, the direction of a line is characterised by a single angle — its inclination to the x-axis, or equivalently its slope. In three-dimensional geometry, direction is more complex: a line in space makes angles with all three coordinate axes simultaneously, and we need three quantities to fully describe its orientation. These are the direction cosines.

Direction Cosines

Let OP be a line through the origin O, making angles α, β, and γ with the positive x-axis, y-axis, and z-axis respectively. The angles α, β, γ are called the direction angles of the line. The direction cosines of the line are:

Direction cosines: $l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma$

The triplet (l, m, n) completely specifies the direction of a line through the origin. Since the direction cosines are components of a unit vector along the line, they satisfy the fundamental identity:

Fundamental identity of direction cosines: $l^2 + m^2 + n^2 = 1$

This identity is the 3D analogue of $\sin^2\theta + \cos^2\theta = 1$ in two dimensions. It is always satisfied by the direction cosines of any line and serves as a verification check.

Note that a line has two directions (forward and backward along it), giving direction cosines (l, m, n) and $(-l, -m, -n)$. Both are valid. Convention typically takes the direction cosines such that $n > 0$ (or $l > 0$ if $n = 0$), but context determines the appropriate choice.

Direction Ratios

The direction ratios of a line are any three numbers a, b, c that are proportional to the direction cosines l, m, n. That is: $l/a = m/b = n/c = k$ for some non-zero constant k.

Direction ratios are not unique — any non-zero scalar multiple of (a, b, c) represents the same direction. They are easier to work with than direction cosines because they avoid the square root in the normalisation.

Given direction ratios a, b, c, the direction cosines are obtained by dividing by the magnitude:

D.C.s from D.R.s: $l = a/\sqrt{(a^2+b^2+c^2)}$, $m = b/\sqrt{(a^2+b^2+c^2)}$, $n = c/\sqrt{(a^2+b^2+c^2)}$

Summary Table

Concept	Description
Direction Angles α, β, γ	Angles the line makes with positive x-, y-, z-axes respectively
Direction Cosines (l,m,n)	$l=\cos\alpha$, $m=\cos\beta$, $n=\cos\gamma$; satisfy $l^2+m^2+n^2=1$
Direction Ratios (a,b,c)	Any numbers proportional to (l,m,n); not unique; $\lvert d.r.s\rvert = \sqrt{(a^2+b^2+c^2)}$
Conversion	$l=a/\sqrt{(a^2+b^2+c^2)}$, $m=b/\ldots$, $n=c/\ldots$
Parallel Lines	Same or negatives of each other direction cosines; d.r.s proportional
Perpendicular Lines	$l_1l_2 + m_1m_2 + n_1n_2 = 0 \iff a_1a_2 + b_1b_2 + c_1c_2 = 0$

Worked Example: Finding Direction Cosines from Direction Ratios

Find the direction cosines of the line with direction ratios (2, −3, 6). Verify $l^2+m^2+n^2=1$.

Step 1: Compute the magnitude: $\sqrt{(2^2+(-3)^2+6^2)} = \sqrt{(4+9+36)} = \sqrt{49} = 7$

Step 2: Direction cosines: $l = 2/7$, $m = -3/7$, $n = 6/7$

Step 3: Verify: $(2/7)^2 + (-3/7)^2 + (6/7)^2 = 4/49 + 9/49 + 36/49 = 49/49 = 1$ ✓

5.2 Direction Cosines of a Line Through Two Points

Given two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in three-dimensional space, the direction ratios of the line PQ are obtained directly from the coordinates of the two points by subtracting the corresponding coordinates:

D.R.s from two points: Direction ratios of PQ: $a = x_2-x_1$, $b = y_2-y_1$, $c = z_2-z_1$

The distance PQ between the two points is:

Distance formula in 3D: $PQ = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]}$

The direction cosines of PQ are:

D.C.s from two points: $l = (x_2-x_1)/PQ$, $m = (y_2-y_1)/PQ$, $n = (z_2-z_1)/PQ$

These are the direction cosines of the line directed from P toward Q. The direction cosines of the line from Q to P would be the negatives of these.

📝 Worked Example: Direction Cosines from Two Points

Find the direction cosines of the line joining A(1, 2, −2) and B(3, −1, 4). Also find the distance AB.

Step 1: Direction ratios: $a = 3-1 = 2$, $b = -1-2 = -3$, $c = 4-(-2) = 6$

Step 2: Distance $AB = \sqrt{(4+9+36)} = \sqrt{49} = 7$

Step 3: Direction cosines: $l = 2/7$, $m = -3/7$, $n = 6/7$

Step 4: Verify: $4/49 + 9/49 + 36/49 = 49/49 = 1$ ✓

The direction cosines of line AB (from A to B) are (2/7, −3/7, 6/7).

💡 Did You Know?

The concept of direction cosines is central to 3D computer graphics and robotics. When ISRO's Chandrayaan-3 lander was descending to the lunar surface in August 2023, its onboard navigation system continuously computed the direction cosines of the local vertical (the direction toward the Moon's centre of mass) relative to its body-fixed coordinate frame. The attitude control thrusters were fired in directions computed from these direction cosines to maintain the correct orientation during the 17-minute powered descent. This is precisely the same mathematics you are studying — expressing directions in 3D space using cosines of angles with coordinate axes.

5.3 Equation of a Line — Vector Form

Having established the language of direction in three-dimensional space, we can now write the equation of a line. The vector form is the most natural and elegant: a line is completely determined by any one point on it and the direction in which it runs.

Let a be the position vector of a fixed point A on the line, and let b be a direction vector along the line (any non-zero vector parallel to the line). Then a point P with position vector r lies on the line if and only if the vector AP = r − a is parallel to b, i.e., r − a = λb for some scalar λ. This gives the vector equation of the line:

Vector equation of a line: $r = a + \lambda b \qquad (\lambda \in \mathbb{R})$

Here, λ is a real parameter. As λ varies over all real numbers, r traces out every point on the line. When λ = 0, r = a (the fixed point). Positive λ gives points in the direction of b beyond A; negative λ gives points in the opposite direction.

Line Through Two Points — Vector Form

If the line passes through two known points A(position vector a) and B(position vector b_point), then the direction vector is b = b_point − a, and the equation is:

Line through two points: $r = a + \lambda(b\_point - a) = (1-\lambda)a + \lambda \cdot b\_point$

When $\lambda = 0$, $r = a$ (point A); when $\lambda = 1$, $r = b\_point$ (point B); when $0 < \lambda < 1$, r is between A and B.

Worked Example: Vector Equation of a Line

(i) Write the vector equation of the line through A(1, 2, 3) with direction (2, −1, 4).

(ii) Write the vector equation of the line through P(1, 0, −1) and Q(3, 2, 1).

Step 1: (i) $a = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$, $b = 2\hat{\imath} - \hat{\jmath} + 4\hat{k}$

Vector equation: $r = (\hat{\imath} + 2\hat{\jmath} + 3\hat{k}) + \lambda(2\hat{\imath} - \hat{\jmath} + 4\hat{k})$

i.e., $r = (1+2\lambda)\hat{\imath} + (2-\lambda)\hat{\jmath} + (3+4\lambda)\hat{k}$

Step 2: (ii) $a = \hat{\imath} + 0\hat{\jmath} - \hat{k}$, direction $b = (3-1)\hat{\imath} + (2-0)\hat{\jmath} + (1-(-1))\hat{k} = 2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$

Simplify direction (divide by 2): $\hat{\imath} + \hat{\jmath} + \hat{k}$

Vector equation: $r = (\hat{\imath} - \hat{k}) + \lambda(\hat{\imath} + \hat{\jmath} + \hat{k})$

i.e., $r = (1+\lambda)\hat{\imath} + \lambda\hat{\jmath} + (-1+\lambda)\hat{k}$

5.4 Equation of a Line — Cartesian Form

The Cartesian (or symmetric) form of the equation of a line expresses the same information as the vector form using explicit x, y, z coordinates. Starting from the vector equation $r = a + \lambda b$ with $a = (x_1, y_1, z_1)$ and $b = (l, m, n)$ (direction cosines or any direction ratios), we set $r = (x, y, z)$ and equate components:

$$x = x_1 + \lambda l, \quad y = y_1 + \lambda m, \quad z = z_1 + \lambda n$$

Solving each equation for λ and equating:

Symmetric (Cartesian) form of a line: $(x - x_1)/l = (y - y_1)/m = (z - z_1)/n = \lambda$

This is called the symmetric form or the standard Cartesian form of a line. If any direction component is zero (say $l = 0$), the corresponding equation becomes $x = x_1$ (a constraint), and the equality is written as $(y-y_1)/m = (z-z_1)/n$ with $x = x_1$ noted separately.

Two-Point Cartesian Form

If the line passes through $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$, the direction ratios are $(x_2-x_1, y_2-y_1, z_2-z_1)$ and the Cartesian equation is:

Two-point Cartesian form: $(x-x_1)/(x_2-x_1) = (y-y_1)/(y_2-y_1) = (z-z_1)/(z_2-z_1)$

Worked Example: Converting Between Vector and Cartesian Forms

(i) Write the Cartesian form of $r = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(\hat{i} + 2\hat{j} - \hat{k})$.

(ii) Convert $(x-1)/2 = (y+2)/3 = (z-3)/(-1)$ to vector form.

Step 1: (i) Point: $(2, -1, 3)$; Direction ratios: $(1, 2, -1)$

Cartesian form: $(x-2)/1 = (y+1)/2 = (z-3)/(-1)$

Step 2: (ii) Point on line: $(1, -2, 3)$; Direction vector: $2\hat{i} + 3\hat{j} - \hat{k}$

Vector form: $r = (\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$

Let's Check

Q.No	Type	Question
1	MCQ	The direction cosines (l, m, n) of a line always satisfy:
2	MCQ	The direction ratios of the line joining (1,2,3) and (4,6,3) are:
3	Fill in Blank	The vector equation of a line through point a with direction b is r = ______.
4	Fill in Blank	If direction ratios are (2, 6, 3), the magnitude is $\sqrt{4+36+9}$ = ______.
5	True/False	Direction ratios are unique for a given line — there is only one set of direction ratios.

6	True/False	In the Cartesian form of a line, all three fractions $(x-x_1)/l = (y-y_1)/m = (z-z_1)/n$ equal the parameter λ.

Answers:

Q.No	Answer
1	b) $l^2+m^2+n^2 = 1$ — the direction cosines form the components of a unit vector.
2	c) (3, 4, 0) — subtract coordinates: (4−1, 6−2, 3−3) = (3, 4, 0).
3	a + λb
4	7 [√49 = 7]
5	False — direction ratios are not unique. Any non-zero scalar multiple of (a,b,c) gives the same direction. Only direction cosines (after normalisation) are unique (up to sign).
6	True — that common ratio equals λ, the parameter that identifies the position of a point on the line.

5.5 Angle Between Two Lines

The angle between two lines in three-dimensional space is defined as the acute angle between their direction vectors (or any vectors parallel to them). Since the two direction vectors may point in either of two opposite directions along each line, we take the absolute value (or consider the acute angle) to ensure we get the angle between 0° and 90°.

If two lines have direction cosines (l_1, m_1, n_1) and (l_2, m_2, n_2), the angle θ between them satisfies:

Angle between lines — direction cosines: $\cos\theta = |l_1l_2 + m_1m_2 + n_1n_2|$

If the lines have direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2), then:

Angle — direction ratios: $\cos\theta = |a_1a_2 + b_1b_2 + c_1c_2| / [\sqrt{(a_1^2+b_1^2+c_1^2)} \cdot \sqrt{(a_2^2+b_2^2+c_2^2)}]$

Special Cases: Parallel and Perpendicular Lines

- Parallel lines: direction vectors are proportional — $a_1/a_2 = b_1/b_2 = c_1/c_2$. Angle = 0°.
- Perpendicular lines: dot product of direction vectors = 0 — $a_1a_2 + b_1b_2 + c_1c_2 = 0$. Angle = 90°.

📝 Worked Example: Angle Between Two Lines

Find the angle between lines with direction ratios (1, 2, 2) and (3, 4, 0). Are they perpendicular?

Step 1: Compute the dot product: $a_1a_2+b_1b_2+c_1c_2 = 1\cdot3 + 2\cdot4 + 2\cdot0 = 3+8+0 = 11$

Step 2: Magnitudes: $|d_1| = \sqrt{(1+4+4)} = \sqrt{9} = 3$, $|d_2| = \sqrt{(9+16+0)} = \sqrt{25} = 5$

Step 3: $\cos \theta = |11|/(3\cdot5) = 11/15$

Step 4: $\theta = \cos^{-1}(11/15) \approx 42.8°$

Since $a_1a_2+b_1b_2+c_1c_2 = 11 \neq 0$, the lines are NOT perpendicular.

📝 Worked Example: Checking Perpendicularity

Show that lines with direction ratios (1, −2, 3) and (2, 3, 1) are perpendicular.

Step 1: Compute dot product: $1\cdot2 + (-2)\cdot3 + 3\cdot1 = 2 - 6 + 3 = -1 \neq 0$

Wait — let us recheck: $(1)(2) + (-2)(3) + (3)(1) = 2 - 6 + 3 = -1 \neq 0$. NOT perpendicular.

Correct example: (1, −2, 3) and (2, 1, 0). Dot product: $2 - 2 + 0 = 0$. ✓ Perpendicular.

5.6 Skew Lines and Shortest Distance — Vector Form

Two lines in three-dimensional space may be in one of three relationships: they may intersect (share a common point), be parallel (same direction, no common point), or be skew (neither intersecting nor parallel — they lie in different planes and never meet). Skew lines are a phenomenon unique to three or more dimensions; in the plane, non-parallel lines always intersect.

Recognising Skew Lines

Two lines are skew if they are neither parallel nor intersecting. To test: if the lines have direction vectors b_1 and b_2 that are not parallel ($b_1 \times b_2 \neq 0$), and the vector connecting points on the two lines is not perpendicular to $b_1 \times b_2$, then the lines are skew.

Shortest Distance Between Skew Lines — Vector Formula

Given two skew lines: L_1: $r = a_1 + \lambda b_1$ and L_2: $r = a_2 + \mu b_2$, the unique common perpendicular connects a point on L_1 to a point on L_2 and is perpendicular to both direction vectors. Its length — the shortest distance — is:

Shortest distance — vector form: $SD = |(a_2 - a_1) \cdot (b_1 \times b_2)| / |b_1 \times b_2|$

Here $(b_1 \times b_2)$ is the cross product of the two direction vectors. The formula computes the component of the vector connecting the two lines $(a_2 - a_1)$ along the direction of the common perpendicular $(b_1 \times b_2)$.

If $SD = 0$, the lines intersect (they are coplanar). If $b_1 \times b_2 = 0$ ($b_1 \parallel b_2$), the lines are parallel, and a different formula applies.

📝 Worked Example: Shortest Distance Between Skew Lines — Vector Form

Find the shortest distance between the lines:

L_1: $r = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$

L_2: $r = (2\hat{i} + 4\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$

Step 1: Identify: $a_1 = \hat{i}+2\hat{j}+3\hat{k}$, $b_1 = 2\hat{i}+3\hat{j}+4\hat{k}$, $a_2 = 2\hat{i}+4\hat{j}+5\hat{k}$, $b_2 = 3\hat{i}+4\hat{j}+5\hat{k}$

Step 2: Compute $a_2-a_1 = (2-1)\hat{i}+(4-2)\hat{j}+(5-3)\hat{k} = \hat{i}+2\hat{j}+2\hat{k}$

Step 3: Compute $b_1\times b_2$ using determinant:

$b_1\times b_2 = |\hat{i}\ \ \hat{j}\ \ \hat{k}\,|$

$|2\ \ 3\ \ 4\,|$

$|3\ \ 4\ \ 5\,|$

$\hat{i}$-component: $3\cdot5 - 4\cdot4 = 15 - 16 = -1$

$\hat{j}$-component: $-(2\cdot5 - 4\cdot3) = -(10-12) = 2$

$\hat{k}$-component: $2\cdot4 - 3\cdot3 = 8 - 9 = -1$

$b_1\times b_2 = -\hat{i} + 2\hat{j} - \hat{k}$

Step 4: $|b_1\times b_2| = \sqrt{(1+4+1)} = \sqrt{6}$

Step 5: $(a_2-a_1)\cdot(b_1\times b_2) = (1)(-1)+(2)(2)+(2)(-1) = -1+4-2 = 1$

Step 6: $SD = |1|/\sqrt{6} = 1/\sqrt{6} = \sqrt{6}/6$

5.7 Shortest Distance Between Skew Lines — Cartesian Form

The Cartesian form of the shortest distance formula is expressed as a determinant, derived by substituting the Cartesian equations of the lines into the vector formula.

For two lines:

L_1: $(x-x_1)/l_1 = (y-y_1)/m_1 = (z-z_1)/n_1$

L_2: $(x-x_2)/l_2 = (y-y_2)/m_2 = (z-z_2)/n_2$

The shortest distance is given by:

SD = Numerator / Denominator, where:

Numerator = |det M| where M is the 3×3 matrix:

$$M = \begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}$$

Shortest distance — Cartesian: $SD = |\det M| / \sqrt{[(m_1n_2-m_2n_1)^2 + (n_1l_2-n_2l_1)^2 + (l_1m_2-l_2m_1)^2]}$

Condition for Lines to Intersect (be Coplanar)

Two lines intersect (and thus lie in a common plane) if and only if the determinant in the numerator is zero — SD = 0:

Condition for intersection: $|x_2-x_1 \quad y_2-y_1 \quad z_2-z_1| = 0$

$|\ l_1 \quad m_1 \quad n_1\ |$

$|\ l_2 \quad m_2 \quad n_2\ |$

📝 Worked Example: Shortest Distance — Cartesian Form

Find the SD between: L_1: $(x-1)/2 = (y-2)/3 = (z-3)/4$ and L_2: $(x-2)/3 = (y-4)/4 = (z-5)/5$

Step 1: $x_1=1, y_1=2, z_1=3$; $l_1=2, m_1=3, n_1=4$; $x_2=2, y_2=4, z_2=5$; $l_2=3, m_2=4, n_2=5$

Step 2: $x_2-x_1=1$, $y_2-y_1=2$, $z_2-z_1=2$

Step 3: Determinant:

$\det = 1(3\cdot5-4\cdot4) - 2(2\cdot5-4\cdot3) + 2(2\cdot4-3\cdot3)$

$= 1(15-16) - 2(10-12) + 2(8-9)$

$= 1(-1) - 2(-2) + 2(-1)$

$= -1 + 4 - 2 = 1$

Step 4: Denominator: $\sqrt{[(3\cdot5-4\cdot4)^2+(4\cdot3-5\cdot2)^2+(2\cdot4-3\cdot3)^2]}$

$= \sqrt{[(-1)^2+(2)^2+(-1)^2]} = \sqrt{6}$

Step 5: $SD = |1|/\sqrt{6} = 1/\sqrt{6}$ ✓ (matches vector form result)

🔍 Analysing Misconceptions: Skew Lines in 3D Behave Like Parallel Lines

✗ MYTH: Skew lines are just a special case of parallel lines — they run in the same direction but are "shifted" apart.

✅ REALITY: Skew lines and parallel lines are fundamentally different. Parallel lines have the same direction (proportional direction vectors) and the shortest distance between them is the constant perpendicular distance. Skew lines have DIFFERENT directions (non-proportional direction vectors) and do not lie in any common plane. The shortest distance between skew lines requires the cross product formula — a completely different calculation. A classic example: the x-axis (y=0, z=0) and the line through (0,1,0) parallel to the z-axis are skew — they point in different directions and never meet.

5.8 Equation of a Plane — Vector Form

Just as a line in space is characterised by a point and a direction, a plane is characterised by a point and a normal direction — a direction perpendicular to the plane. Every vector lying in the plane is perpendicular to the normal vector. This geometric insight gives the vector equation of the plane.

Let a be the position vector of a fixed point A on the plane, and let $\hat{n}$ be the unit normal to the plane. A point P with position vector r lies on the plane if and only if the vector AP = r − a is perpendicular to $\hat{n}$, i.e., $(r - a)\cdot\hat{n} = 0$. This gives:

Vector equation of plane — unit normal form: $r \cdot \hat{n} = a \cdot \hat{n} = d$

where $d = a\cdot\hat{n}$ is the perpendicular distance from the origin to the plane. Alternatively, for any normal vector n (not necessarily unit):

Vector equation of plane — general normal: $r \cdot n = a \cdot n$

Vector Equation Through Three Points

If a plane passes through three non-collinear points A, B, C with position vectors a, b, c, the two vectors AB = b−a and AC = c−a lie in the plane. The normal to the plane is $n = (b-a) \times (c-a)$, and the equation is:

Plane through three points — vector: $(r - a) \cdot [(b-a) \times (c-a)] = 0$

Worked Example: Vector Equation of a Plane

Find the vector equation of the plane through A(1,2,3) with normal $n = 2\hat{i}-\hat{j}+3\hat{k}$.

Step 1: $a = \hat{i}+2\hat{j}+3\hat{k}$, $n = 2\hat{i}-\hat{j}+3\hat{k}$

Step 2: $d = a\cdot n = 1\cdot 2 + 2\cdot(-1) + 3\cdot 3 = 2-2+9 = 9$

Step 3: Vector equation: $r\cdot(2\hat{i}-\hat{j}+3\hat{k}) = 9$

In Cartesian form: $2x - y + 3z = 9$

5.9 Equation of a Plane — Cartesian Forms

The Cartesian equation of a plane is obtained from the vector form by substituting $r = x\hat{i} + y\hat{j} + z\hat{k}$ and $n = a\hat{i} + b\hat{j} + c\hat{k}$. The result is the general equation of a plane:

General Cartesian equation of a plane: $ax + by + cz + d = 0$

Here (a, b, c) are the direction ratios of the normal to the plane, and d is a constant. This is the most general form — every plane in three dimensions has an equation of this form (for appropriate a, b, c, d with a, b, c not all zero).

Intercept Form

If the plane makes intercepts p, q, r on the x-, y-, and z-axes respectively, its equation is:

Intercept form: $x/p + y/q + z/r = 1$

This is convenient when intercepts are given directly. Setting $y = z = 0$ gives $x = p$ (the x-intercept), and so on.

Plane Through Three Points — Cartesian

To find the equation of the plane through three non-collinear points (x_1,y_1,z_1), (x_2,y_2,z_2), (x_3,y_3,z_3), expand the determinant condition that the vectors from the first point to the other two lie in the plane:

$$\det \begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

📝 Worked Example: Plane Through Three Points

Find the equation of the plane through (1,1,0), (1,2,1), and (−2,2,−1).

Step 1: Vectors from first point: $v_1 = (0,1,1)$, $v_2 = (-3,1,-1)$

Step 2: Normal $n = v_1 \times v_2$:

$\hat{\imath}$: $1\cdot(-1)-1\cdot1 = -2$

$\hat{\jmath}$: $-(0\cdot(-1)-1\cdot(-3)) = -(3) = -3$

$\hat{k}$: $0\cdot1-1\cdot(-3) = 3$

$n = -2\hat{\imath} - 3\hat{\jmath} + 3\hat{k}$ (or equivalently $2\hat{\imath} + 3\hat{\jmath} - 3\hat{k}$)

Step 3: Using point (1,1,0): $2(x-1) + 3(y-1) - 3(z-0) = 0$

$2x + 3y - 3z - 5 = 0$

Step 4: Verify with (1,2,1): $2+6-3-5 = 0$ ✓; with (−2,2,−1): $-4+6+3-5 = 0$ ✓

Worked Example: Intercept Form and Normal Form

A plane makes intercepts 3, −4, 6 on the coordinate axes. Find its equation.

Step 1: Intercept form: $x/3 + y/(-4) + z/6 = 1$

Step 2: Multiply through by 12: $4x - 3y + 2z = 12$

Step 3: General form: $4x - 3y + 2z - 12 = 0$

Normal vector: (4, −3, 2). This plane cuts the x-axis at (3,0,0), y-axis at (0,−4,0), z-axis at (0,0,6).

Let's Check

Q.No	Type	Question
1	MCQ	The angle θ between two lines with direction ratios (a_1,b_1,c_1) and (a_2,b_2,c_2) satisfies cos θ = :
2	MCQ	The shortest distance between skew lines L_1: $r = a_1+\lambda b_1$ and L_2: $r = a_2+\mu b_2$ is:
3	Fill in Blank	Two lines are perpendicular if their direction ratios satisfy $a_1a_2+b_1b_2+c_1c_2$ = ______.
4	Fill in Blank	The general equation of a plane is $ax + by + cz + d = 0$, where (a, b, c) are the direction ratios of the ______ to the plane.
5	True/False	Two skew lines always have a unique common perpendicular whose length equals the shortest distance.

6	True/False	The intercept form x/p + y/q + z/r = 1 is valid even when one intercept is zero.

Answers:

Q.No	Answer
1	b) $\lvert a_1a_2+b_1b_2+c_1c_2\rvert / [\sqrt{(a_1^2+b_1^2+c_1^2)}\cdot\sqrt{(a_2^2+b_2^2+c_2^2)}]$
2	c) $\lvert(a_2-a_1)\cdot(b_1\times b_2)\rvert / \lvert b_1\times b_2\rvert$
3	0
4	normal vector
5	True — for any two skew lines there exists a unique common perpendicular, and its length is the shortest distance.
6	False — if any intercept is zero, the plane passes through the origin (that axis), and intercept form becomes undefined (division by zero). Use the general form ax+by+cz+d=0 instead.

5.10 Angle Between Two Planes

The angle between two planes is defined as the acute angle between their normal vectors. Since any plane has two opposite normals (n and −n), we take the absolute value to ensure we get the acute dihedral angle between the planes.

For planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ with normals $n_1 = (a_1,b_1,c_1)$ and $n_2 = (a_2,b_2,c_2)$:

Angle between planes: $\cos\theta = |n_1 \cdot n_2| / (|n_1||n_2|) = |a_1a_2+b_1b_2+c_1c_2| / [\sqrt{(a_1^2+b_1^2+c_1^2)} \cdot \sqrt{(a_2^2+b_2^2+c_2^2)}]$

Special Cases

- Parallel planes: normal vectors proportional — $a_1/a_2 = b_1/b_2 = c_1/c_2$. The planes never intersect.
- Perpendicular planes: $a_1a_2 + b_1b_2 + c_1c_2 = 0$. The normals are perpendicular, so the planes meet at 90°.

Distance from a Point to a Plane

The perpendicular distance from a point $P(x_0, y_0, z_0)$ to the plane $ax + by + cz + d = 0$ is:

Point-to-plane distance: $\text{distance} = |ax_0 + by_0 + cz_0 + d| / \sqrt{(a^2 + b^2 + c^2)}$

The distance between two parallel planes $ax+by+cz+d_1 = 0$ and $ax+by+cz+d_2 = 0$ is:

Distance between parallel planes: $\text{distance} = |d_1 - d_2| / \sqrt{(a^2 + b^2 + c^2)}$

📝 Worked Example: Angle Between Planes and Point-to-Plane Distance

Find: (i) the angle between $2x-y+2z=5$ and $3x+2y-6z=7$; (ii) distance from $(2,3,-1)$ to $2x-y+2z=5$.

Step 1: (i) $n_1=(2,-1,2)$, $n_2=(3,2,-6)$

$n_1 \cdot n_2 = 6-2-12 = -8$

$|n_1|=\sqrt{(4+1+4)}=3$, $|n_2|=\sqrt{(9+4+36)}=7$

$\cos\theta = |-8|/(3\cdot 7) = 8/21$

$\theta = \cos^{-1}(8/21) \approx 67.6°$

Step 2: (ii) $d = |2(2)-1(3)+2(-1)-5|/\sqrt{(4+1+4)} = |4-3-2-5|/3 = |-6|/3 = 2$

Distance from $(2,3,-1)$ to the plane $2x-y+2z=5$ is 2 units.

5.11 Angle Between a Line and a Plane

The angle between a line and a plane is the complement of the angle between the line's direction vector and the plane's normal vector. If φ is the angle between the direction vector b of the line and the normal n of the plane, then the angle θ between the line and the plane satisfies $\theta = 90° - \varphi$, and therefore:

Angle between line and plane: $\sin\theta = |b \cdot n| / (|b| \cdot |n|) = |a \cdot l + b \cdot m + c \cdot n| / [\sqrt{(a^2+b^2+c^2)} \cdot \sqrt{(l^2+m^2+n^2)}]$

where (l, m, n) are the direction ratios of the line and (a, b, c) are the direction ratios of the normal to the plane.

Special Cases

- Line parallel to plane: $b \cdot n = 0$ (direction vector perpendicular to normal, i.e., direction vector lies in plane direction) — angle between line and plane is 0°.
- Line perpendicular to plane: b is parallel to n (direction ratios proportional) — angle is 90°.
- Line lies in the plane: $b \cdot n = 0$ AND point on line satisfies the plane equation.

Worked Example: Angle Between Line and Plane

Find the angle between the line $(x-1)/1 = (y-2)/2 = (z-3)/(-1)$ and the plane $2x+y-z=5$.

Step 1: Direction ratios of line: $b = (1,2,-1)$

Step 2: Normal to plane: $n = (2,1,-1)$

Step 3: $b \cdot n = 1 \cdot 2 + 2 \cdot 1 + (-1)(-1) = 2+2+1 = 5$

Step 4: $|b| = \sqrt{(1+4+1)} = \sqrt{6}$, $|n| = \sqrt{(4+1+1)} = \sqrt{6}$

Step 5: $\sin \theta = |5|/(\sqrt{6} \cdot \sqrt{6}) = 5/6$

Step 6: $\theta = \sin^{-1}(5/6) \approx 56.4°$

5.12 Coplanar Lines

Two lines in three-dimensional space are coplanar if they lie in a common plane. This happens when the lines either intersect (they share a point, and any two intersecting lines determine a plane) or are parallel (parallel lines always lie in a common plane). Skew lines are never coplanar.

Condition for Coplanarity — Vector Form

Two lines L_1: $r = a_1 + \lambda b_1$ and L_2: $r = a_2 + \mu b_2$ are coplanar if and only if the vector $a_2 - a_1$ lies in the plane spanned by b_1 and b_2. The condition is:

Coplanarity condition — vector: $(a_2 - a_1) \cdot (b_1 \times b_2) = 0$

This is equivalent to saying SD = 0 (the shortest distance between the lines is zero — they meet or are parallel). If this condition holds and the lines are not parallel ($b_1 \times b_2 \neq 0$), the lines intersect at a unique point.

Condition for Coplanarity — Cartesian Form

For lines L_1: $(x-x_1)/l_1 = (y-y_1)/m_1 = (z-z_1)/n_1$ and L_2: $(x-x_2)/l_2 = (y-y_2)/m_2 = (z-z_2)/n_2$, the coplanarity condition is:

$$\det \begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

Finding the Equation of the Plane Containing Two Coplanar Lines

If the two lines are coplanar, the plane containing them has normal vector $n = b_1 \times b_2$ and passes through the point a_1 (on L_1). The equation is: $(r - a_1) \cdot (b_1 \times b_2) = 0$.

Worked Example: Testing Coplanarity and Finding the Common Plane

Test if the lines are coplanar and find the equation of their common plane:

L_1: $(x-1)/2 = (y-2)/3 = (z-3)/4$

L_2: $(x-2)/3 = (y-3)/4 = (z-4)/5$

Step 1: $x_2-x_1=1$, $y_2-y_1=1$, $z_2-z_1=1$; $l_1=2, m_1=3, n_1=4$; $l_2=3, m_2=4, n_2=5$

Step 2: Determinant test:

$\det = 1(3\cdot5-4\cdot4) - 1(2\cdot5-4\cdot3) + 1(2\cdot4-3\cdot3)$

$= 1(15-16) - 1(10-12) + 1(8-9)$

$= -1 + 2 - 1 = 0$ ✓ Lines ARE coplanar!

Step 3: Normal $n = b_1 \times b_2$:

$n = (3\cdot5-4\cdot4)\hat{i} - (2\cdot5-4\cdot3)\hat{j} + (2\cdot4-3\cdot3)\hat{k}$

$= -\hat{i} + 2\hat{j} - \hat{k}$ (or $\hat{i} - 2\hat{j} + \hat{k}$)

Step 4: Plane through (1,2,3) with normal $\hat{i}-2\hat{j}+\hat{k}$:

$1(x-1) - 2(y-2) + 1(z-3) = 0$

$x - 2y + z - 1 + 4 - 3 = 0$

$x - 2y + z = 0$

Step 5: Verify with (2,3,4): $2-6+4 = 0$ ✓

5.13 Intersection of a Line and a Plane

A line in three-dimensional space, unless it is parallel to a plane or lies entirely within it, intersects the plane at exactly one point. Finding this point of intersection is a fundamental problem in three-dimensional geometry with direct applications in computer graphics (ray tracing, collision detection), robotics (workspace boundary analysis), and architectural design (finding where a beam meets a wall).

Method: Substituting the Line into the Plane

To find the intersection of the line $(x-x_1)/l = (y-y_1)/m = (z-z_1)/n = \lambda$ with the plane $ax+by+cz+d = 0$:

- Step 1: Write the general point on the line as $(x_1+l\lambda, y_1+m\lambda, z_1+n\lambda)$.
- Step 2: Substitute into the plane equation: $a(x_1+l\lambda) + b(y_1+m\lambda) + c(z_1+n\lambda) + d = 0$.
- Step 3: Solve for λ.
- Step 4: Substitute back to find the coordinates of the intersection point.

The line is parallel to the plane if $al+bm+cn = 0$. In this case, if the point (x_1,y_1,z_1) does not satisfy the plane equation, there is no intersection (line parallel to but outside plane). If it does satisfy the equation, the line lies entirely in the plane.

Foot of the Perpendicular from a Point to a Plane

The foot of the perpendicular from a point $P(x_0, y_0, z_0)$ to the plane $ax+by+cz+d = 0$ is found by treating the perpendicular as a line through P in the direction of the normal (a, b, c):

Foot of perpendicular: $(x-x_0)/a = (y-y_0)/b = (z-z_0)/c = -(ax_0+by_0+cz_0+d)/(a^2+b^2+c^2) = \lambda$

The foot is the point $(x_0+a\lambda, y_0+b\lambda, z_0+c\lambda)$ where $\lambda = -(ax_0+by_0+cz_0+d)/(a^2+b^2+c^2)$.

📝 Worked Example: Intersection of Line and Plane

Find where the line $(x-1)/2 = (y-2)/3 = (z+1)/(-1)$ meets the plane $x+y+z = 4$.

Step 1: General point on line: $x = 1+2\lambda$, $y = 2+3\lambda$, $z = -1-\lambda$

Step 2: Substitute into plane $x+y+z = 4$:

$(1+2\lambda) + (2+3\lambda) + (-1-\lambda) = 4$

$2 + 4\lambda = 4$

$4\lambda = 2 \implies \lambda = 1/2$

Step 3: Point of intersection:

$x = 1+2(1/2) = 2, \quad y = 2+3(1/2) = 7/2, \quad z = -1-(1/2) = -3/2$

Intersection point: $(2, 7/2, -3/2)$

Step 4: Verify: $2 + 7/2 + (-3/2) = 2 + 2 = 4$ ✓

📝 Worked Example: Foot of Perpendicular from Point to Plane

Find the foot of the perpendicular from $A(1, 3, 4)$ to the plane $2x - y + 2z = 9$. Also find the distance.

Step 1: Line through A perpendicular to plane has direction $(2,-1,2)$. Parametric:

$x = 1+2\lambda$, $y = 3-\lambda$, $z = 4+2\lambda$

Step 2: Substitute into $2x-y+2z=9$:

$2(1+2\lambda) - (3-\lambda) + 2(4+2\lambda) = 9$

$2+4\lambda-3+\lambda+8+4\lambda = 9$

$7 + 9\lambda = 9 \implies \lambda = 2/9$

Step 3: Foot of perpendicular:

$x = 1+4/9 = 13/9, \quad y = 3-2/9 = 25/9, \quad z = 4+4/9 = 40/9$

Step 4: Distance $= |2(1)-1(3)+2(4)-9|/\sqrt{(4+1+4)} = |2-3+8-9|/3 = |-2|/3 = 2/3$

Did You Know?

Three-dimensional geometry forms the mathematical backbone of all modern computer graphics and game engines. When you play a game like BGMI (Battlegrounds Mobile India) — one of the most downloaded games in India with over 100 million downloads — every collision detection, bullet trajectory, and camera frustum calculation uses precisely the formulas studied in this module: equations of lines (bullet paths), planes (walls, floors, terrain surfaces), intersection of line and plane (where the bullet hits the surface), and distance from point to plane (checking if an object is inside or outside a room). The 3D rendering pipeline in Unreal Engine and Unity — used by Indian game studios like Nazara Technologies and GameDuell — executes these geometric computations millions of times per second.

Let's Check

Q.No	Type	Question
1	MCQ	The angle θ between two planes $a_1x+b_1y+c_1z=d_1$ and $a_2x+b_2y+c_2z=d_2$ is found using:
2	MCQ	Two lines are coplanar if the determinant test gives:
3	Fill in Blank	The distance from point (x_0,y_0,z_0) to plane $ax+by+cz+d=0$ is ______.
4	Fill in Blank	The angle between a line (direction b) and a plane (normal n) uses the formula sin θ = ______.
5	True/False	If two lines intersect, they are necessarily coplanar.

6	True/False	The foot of the perpendicular from a point to a plane lies on the line through the point in the direction of the plane's normal vector.

Answers:

Q.No	Answer
1	b) $\cos\theta = \lvert n_1 \cdot n_2 \rvert / (\lvert n_1 \rvert \lvert n_2 \rvert)$ — angle between their normal vectors.
2	a) 0 — the determinant condition equals zero for coplanar lines (SD = 0).
3	$\lvert ax_0+by_0+cz_0+d \rvert / \sqrt{(a^2+b^2+c^2)}$
4	$\lvert b \cdot n \rvert / (\lvert b \rvert \lvert n \rvert)$
5	True — two intersecting lines always lie in a common plane (they determine the plane).
6	True — the perpendicular from a point to a plane is in the direction of the normal vector n. The foot is found by tracing this line to the plane.

📘 Case Study: Three-Dimensional Geometry in Indian Railway Bridge Engineering — Konkan Railway

Background:

The Konkan Railway, stretching 756 km along India's western coastline from Mumbai to Mangaluru, is one of the most engineering-demanding rail projects ever built. Commissioned in 1998 and managed by Konkan Railway Corporation Ltd (KRCL), the route traverses the Western Ghats — featuring 2,000 bridges, 93 tunnels, and terrain rising and falling sharply over short distances. Every structural element of this engineering feat required precise three-dimensional geometry, including the concepts studied in this module.

Direction Cosines in Bridge Alignment:

When the Konkan Railway passes through a mountain range, the track must change direction to follow the valley while maintaining safe gradients. At each such change of direction, the structural engineers compute the direction cosines of the incoming and outgoing track segments. The angle between two track sections — a vital safety parameter — is computed using the dot product formula: $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$. Indian Railway standards (RDSO guidelines) require this angle to not exceed 5° per transition section for safe high-speed operation.

The Panval Viaduct — Lines and Planes in Structural Analysis:

The Panval River Viaduct (near Ratnagiri, Maharashtra) is the tallest railway bridge in India at 64 metres above the river bed. Each of its concrete box-girder spans must remain within a specified plane — any deviation (called "out-of-plumb" error) compromises structural integrity. Engineers verify this using the equation of a plane through three reference survey points. The perpendicular distance from any fourth survey point to this reference plane measures the out-of-plumb error. Using the formula $d = |ax_1+by_1+cz_1+d|/\sqrt{a^2+b^2+c^2}$ studied in Sub-unit 5.11, any deviation greater than 5 mm triggers a structural review.

Tunnel Alignment — Intersection of Lines with Planes:

The Konkan Railway has the Karbude Tunnel (6.5 km — the longest in India at the time of its construction). The tunnel must break through a series of rock layers — each geological stratum is a plane with a known equation (determined by geological surveys giving normal direction and a depth reference point). The tunnel's centreline is the line, and the geotechnical team computes the exact point where the tunnel meets each rock stratum using the intersection formula from Sub-unit 5.13. This determines where drill patterns and blast sequences must change — a direct application of $(x-x_1)/l = (y-y_1)/m = (z-z_1)/n$ substituted into $ax+by+cz+d = 0$.

Skew Lines and Cable Management:

In the major cable-stayed bridges on the Konkan route, the structural cables run in different planes and at different angles — many are skew lines. The shortest distance between two skew cables determines the minimum required clearance to prevent vibration-induced contact under wind loading. KRCL structural engineers use the formula $SD = |(a_2-a_1)\cdot(b_1\times b_2)|/|b_1\times b_2|$ to compute this clearance for every pair of cables. The Indian standard IS:9595 requires a minimum clearance of 300 mm between any two structural cables for safety and maintenance access.

(Source: Konkan Railway Corporation Ltd project documentation; RDSO track geometry standards; Indian Standard IS:9595.)

Module Summary

Module 5 has developed a comprehensive and rigorous treatment of three-dimensional geometry, completing the Fundamentals of Mathematics course. This module brought together the algebraic tools of linear algebra (Module 1), the vector operations of Module 2, and the analytical thinking of Modules 3 and 4, applying them all to the rich geometric environment of three-dimensional space.

The module opened with direction cosines and direction ratios — the fundamental language of orientation in 3D space. Direction cosines (l, m, n) are the cosines of the angles the line makes with the x-, y-, and z-axes, always satisfying $l^2+m^2+n^2=1$. Direction ratios (a, b, c) are any proportional set. Converting between them requires dividing by the magnitude $\sqrt{(a^2+b^2+c^2)}$.

Lines in three-dimensional space were studied in two equivalent representations. The vector form $r = a + \lambda b$ is compact, geometric, and easy to manipulate. The Cartesian symmetric form $(x-x_1)/l = (y-y_1)/m = (z-z_1)/n$ connects directly to coordinates. The angle between two lines uses the dot product of direction vectors; lines are parallel when direction ratios are proportional, and perpendicular when the dot product is zero.

Three-dimensional space introduces the phenomenon of skew lines — lines that neither intersect nor are parallel, having no common point and lying in different planes. The shortest distance between skew lines requires the cross product: $SD = |(a_2-a_1)\cdot(b_1\times b_2)|/|b_1\times b_2|$. This determinant structure is mirrored in the Cartesian form. When SD = 0, the lines are coplanar (either intersecting or parallel).

Planes were treated in multiple forms: the vector point-normal form $r\cdot n = d$; the general Cartesian form $ax+by+cz+d=0$; the intercept form $x/p+y/q+z/r=1$; and the three-point determinant form. The angle between two planes equals the angle between their normals. The perpendicular distance from a point to a plane uses the clean formula $d = |ax_0+by_0+cz_0+d|/\sqrt{(a^2+b^2+c^2)}$.

The final topics synthesised all earlier work. Coplanarity of two lines requires the vanishing of a 3×3 determinant — this determines whether two lines lie in a common plane (and if so, provides the equation of that plane). The intersection of a line and a plane is found by parametrising the line and substituting into the plane equation — a direct algebraic technique with immediate practical applications in engineering, graphics, and robotics.

Glossary

Term	Definition
Direction Cosines	The cosines of the angles a line makes with the positive x, y, z axes: $(l,m,n) = (\cos\alpha, \cos\beta, \cos\gamma)$. Satisfy $l^2+m^2+n^2=1$.
Direction Ratios	Any three numbers (a,b,c) proportional to the direction cosines. Not unique — any scalar multiple is equivalent.
Vector Equation of a Line	$r = a + \lambda b$, where a is a point on the line and b is the direction vector. λ is the real parameter.
Symmetric Form of a Line	$(x-x_1)/l = (y-y_1)/m = (z-z_1)/n$. The Cartesian representation of a line through (x_1,y_1,z_1) with direction (l,m,n).
Skew Lines	Lines in 3D space that are neither parallel nor intersecting. They do not lie in any common plane.
Shortest Distance	The length of the unique common perpendicular to two skew lines. Formula: $\lvert(a_2-a_1)\cdot(b_1\times b_2)\rvert / \lvert b_1\times b_2\rvert$.
Equation of a Plane	$ax+by+cz+d=0$ (general Cartesian form), where (a,b,c) is the normal vector. Also: $r\cdot n=d$ in vector form.
Coplanar Lines	Two lines lying in a common plane. Condition: the determinant $\lvert(a_2-a_1, b_1, b_2)\rvert = 0$ (scalar triple product = 0).
Angle Between Planes	The acute angle between the normal vectors of the two planes: $\cos\theta = \lvert n_1\cdot n_2\rvert/(\lvert n_1\rvert\lvert n_2\rvert)$.
Foot of Perpendicular	The point on a plane closest to a given external point. Found by tracing the normal from the point to the plane.

Self Assessment Questions

Section A: Multiple Choice Questions (MCQ)

Choose the most appropriate answer for each question.

1. The direction cosines (l, m, n) of a line always satisfy:

   a) $l+m+n=1$

   b) $l^2+m^2+n^2=1$

   c) $l^2+m^2+n^2=0$

   d) $l\cdot m\cdot n=1$

2. The direction ratios of the line joining $(2,3,-1)$ and $(4,5,1)$ are:

   a) (2,2,2)

   b) (6,8,0)

   c) (2,2,2) — simplified to (1,1,1)

   d) (1,1,1) directly

3. The vector equation of a line through point a with direction b is:

   a) $r = b + \lambda a$

   b) $r\cdot a = b$

   c) $r = a + \lambda b$

   d) $r\times b = a$

4. Two lines with direction ratios (1,2,3) and (2,4,6) are:

   a) Perpendicular

   b) Skew

   c) Parallel

   d) Intersecting at origin

5. The condition for two lines to be perpendicular using direction ratios is:

   a) $a_1/a_2 = b_1/b_2 = c_1/c_2$

   b) $a_1a_2+b_1b_2+c_1c_2=0$

c) $a_1a_2+b_1b_2+c_1c_2=1$

d) $a_1+b_1+c_1=0$

6. The shortest distance between skew lines L_1: $r=a_1+\lambda b_1$ and L_2: $r=a_2+\mu b_2$ is:

a) $|b_1\times b_2|/|(a_2-a_1)|$

b) $|(a_2-a_1)\cdot(b_1\times b_2)|/|b_1\times b_2|$

c) $(a_2-a_1)/(b_1\cdot b_2)$

d) $|a_2-a_1|$

7. Two skew lines are coplanar if and only if:

a) Their directions are parallel

b) The determinant of the 3×3 matrix equals zero

c) Their shortest distance is 1

d) They have the same direction cosines

8. The general equation of a plane is $ax+by+cz+d=0$. The normal vector is:

a) (d,d,d)

b) (x,y,z)

c) (a,b,c)

d) (a+b,b+c,c+a)

9. The distance from origin to the plane $2x-3y+6z=7$ is:

a) 7

b) 1

c) 7/7=1

d) $7/\sqrt{(4+9+36)} = 1$

10. The angle between planes $2x+y-2z=3$ and $x-2y+2z=1$ is found using:

a) The angle between the planes' x-intercepts

b) $\cos\theta = |n_1\cdot n_2|/(|n_1||n_2|)$

c) $\tan\theta = |n_1||n_2|$

d) The distance formula

11. The intercept form of the equation of a plane is:

a) $ax+by+cz=1$

b) $x/a + y/b + z/c = 0$

c) $x/p + y/q + z/r = 1$

d) $px+qy+rz=0$

12. The angle between a line (direction b) and a plane (normal n) satisfies:

a) $\cos \theta = |b \cdot n|/(|b||n|)$

b) $\sin \theta = |b \cdot n|/(|b||n|)$

c) $\tan \theta = |b \cdot n|/(|b||n|)$

d) $\theta = \cos^{-1}(b \times n)$

13. To find the intersection of a line and a plane, we:

a) Set the line equal to the plane equation

b) Parametrise the line and substitute into the plane

c) Find the cross product of line and plane

d) Compute the shortest distance

14. Two planes are parallel if their normals (a_1,b_1,c_1) and (a_2,b_2,c_2) satisfy:

a) $a_1a_2+b_1b_2+c_1c_2=0$

b) $a_1/a_2=b_1/b_2=c_1/c_2$

c) $a_1+a_2=b_1+b_2=c_1+c_2$

d) $(a_1,b_1,c_1)=(0,0,0)$

15. The foot of perpendicular from a point P to a plane lies on:

a) A line through P parallel to the plane

b) The intersection of two planes

c) The line through P in the direction of the plane's normal

d) The z-axis

16. Direction cosines (l,m,n) of a line with direction ratios (3,4,0) are:

a) (3,4,0)

b) (3/5, 4/5, 0)

c) (3/7, 4/7, 0)

d) (9/25, 16/25, 0)

17. The distance between parallel planes 2x+y+2z=3 and 2x+y+2z=9 is:

a) 6

b) 3

c) 2

d) 9

18. For lines to be coplanar (using vector form), the condition is:

a) $b_1 \cdot b_2 = 0$

b) $(a_2 - a_1) \cdot (b_1 \times b_2) = 0$

c) $a_1 = a_2$

d) $b_1 = b_2$

19. The symmetric form of a line through (1,−2,3) with direction ratios (2,3,−1) is:

a) (x+1)/2=(y−2)/3=(z+3)/(−1)

b) (x−1)/2=(y+2)/3=(z−3)/(−1)

c) (x−2)/1=(y−3)/(−2)=(z+1)/3

d) x/2=y/3=z/(−1)

20. The equation of a plane through three points A, B, C (non-collinear) uses:

a) The average of three coordinates

b) A 3×3 determinant expansion

c) The sum of position vectors

d) The midpoint formula

Section B: True / False

State whether each statement is True or False.

1. In three-dimensional space, two non-parallel lines always intersect.

2. The normal vector to the plane $3x - 2y + z = 5$ has direction ratios $(3, -2, 1)$.

3. Skew lines can have a shortest distance of zero.

4. The angle between a line and a plane can be computed as the complement of the angle between the line's direction vector and the plane's normal.

5. Two intersecting lines are always coplanar.

Section C: Short Answer / Problem Solving

Answer each question with full working, clearly showing all steps.

1. Two lines are given as L_1: $r = (2\hat{i}+\hat{j}-\hat{k}) + \lambda(\hat{i}+2\hat{j}+2\hat{k})$ and L_2: $r = (3\hat{i}+2\hat{j}-2\hat{k}) + \mu(2\hat{i}-\hat{j}+\hat{k})$. Find: (i) the angle between the lines, (ii) whether they are coplanar or skew, (iii) if skew, find the shortest distance.

2. Find the equation of the plane passing through the three points $A(1,1,-1)$, $B(2,0,2)$, and $C(0,2,-1)$. Also find the perpendicular distance from the point $D(3,1,2)$ to this plane.

3. Find the point of intersection of the line $(x-1)/3 = (y-2)/1 = (z-3)/(-2)$ with the plane $2x-y+z=5$. Also find the angle this line makes with the plane.

4. Prove that the lines L_1: $(x-1)/1 = (y-2)/2 = (z-3)/3$ and L_2: $(x-2)/2 = (y-3)/3 = (z-4)/4$ are coplanar. Find the equation of the plane that contains both lines.

5. A plane makes intercepts a, b, c on the coordinate axes. If the plane passes through (1, 2, 3) and is perpendicular to the line with direction ratios (2, 3, 1), find the equation of the plane. Also find the angle this plane makes with the xy-plane ($z=0$).

Self Assessment — Answer Keys

Section A — MCQ Answer Key

Q.No	Answer	Option Text
1	**B**	b) $l^2+m^2+n^2=1$
2	**A**	a) (2,2,2)
3	**C**	c) r = a + λb
4	**C**	c) Parallel
5	**B**	b) $a_1a_2+b_1b_2+c_1c_2=0$
6	**B**	b) $\|(a_2-a_1)\cdot(b_1\times b_2)\|/\|b_1\times b_2\|$
7	**B**	b) The determinant of the 3×3 matrix equals zero
8	**C**	c) (a,b,c)
9	**D**	d) $7/\sqrt{(4+9+36)} = 1$
10	**B**	b) $\cos\theta = \|n_1\cdot n_2\|/(\|n_1\|\|n_2\|)$
11	**C**	c) $x/p + y/q + z/r = 1$
12	**B**	b) $\sin\theta = \|b\cdot n\|/(\|b\|\|n\|)$
13	**B**	b) Parametrise the line and substitute into the plane
14	**B**	b) $a_1/a_2=b_1/b_2=c_1/c_2$
15	**C**	c) The line through P in the direction of the plane's normal
16	**B**	b) (3/5, 4/5, 0)
17	**C**	c) 2
18	**B**	b) $(a_2-a_1)\cdot(b_1\times b_2)=0$

19	**B**	b) $(x-1)/2=(y+2)/3=(z-3)/(-1)$
20	**B**	b) A 3×3 determinant expansion

Section B — True / False Answer Key

Q.No	Answer	Notes
1	**False**	False — in 3D, two non-parallel lines may be skew (neither parallel nor intersecting). Intersection is not guaranteed.
2	**True**	Correct.
3	**False**	False — skew lines by definition never meet (no common point). SD = 0 would mean the lines intersect, making them coplanar, not skew.
4	**True**	Correct.
5	**True**	Correct.

Section C — Short Answer Model Solutions

Q.No	Model Solution
1	(i) $b_1=(1,2,2)$, $b_2=(2,-1,1)$. $b_1\cdot b_2=2-2+2=2$. $\lvert b_1\rvert=3$, $\lvert b_2\rvert=\sqrt{6}$. $\cos\theta=2/(3\sqrt{6})$. $\theta=\cos^{-1}(2/(3\sqrt{6}))\approx 74.2°$. (ii) Coplanarity test: $a_2-a_1=(3-2)\hat{\imath}+(2-1)\hat{\jmath}+(-2+1)\hat{k}=\hat{\imath}+\hat{\jmath}-\hat{k}$. $b_1\times b_2$: $\hat{\imath}(2\cdot 1-2\cdot(-1))-\hat{\jmath}(1\cdot 1-2\cdot 2)+\hat{k}(1\cdot(-1)-2\cdot 2)=\hat{\imath}(4)-\hat{\jmath}(-3)+\hat{k}(-5)=4\hat{\imath}+3\hat{\jmath}-5\hat{k}$. $(a_2-a_1)\cdot(b_1\times b_2)=4+3+5=12\neq 0$. Lines are SKEW. (iii) $SD=\lvert 12\rvert/\sqrt{(16+9+25)}=12/\sqrt{50}=12/(5\sqrt{2})=6\sqrt{2}/5$.

2	Vectors: AB=(1,−1,3), AC=(−1,1,0). Normal n=AB×AC: $\hat{\imath}(-1\cdot0-3\cdot1)-\hat{\jmath}(1\cdot0-3\cdot(-1))+\hat{k}(1\cdot1-(-1)(-1))=-3\hat{\imath}-3\hat{\jmath}+0\hat{k}$. Use (−3,−3,0) or (1,1,0). Through A(1,1,−1): (x−1)+(y−1)=0 → x+y=2. Verify B(2,0,2): 2+0=2 ✓; C(0,2,−1): 0+2=2 ✓. Distance from D(3,1,2): $\|3+1-2\|/\sqrt{2} = 2/\sqrt{2} = \sqrt{2}$.
3	Parametric point: (1+3λ, 2+λ, 3−2λ). Substitute: 2(1+3λ)−(2+λ)+(3−2λ)=5. 2+6λ−2−λ+3−2λ=5. 3+3λ=5 → λ=2/3. Point: (1+2, 2+2/3, 3−4/3)=(3, 8/3, 5/3). Angle with plane: b=(3,1,−2), n=(2,−1,1). sin $\theta=\|b\cdot n\|/(\|b\|\|n\|)=\|6-1-2\|/(\sqrt{14}\cdot\sqrt{6})=3/\sqrt{84}=3/(2\sqrt{21})=\sqrt{21}/14$. $\theta=\sin^{-1}(3/\sqrt{84})\approx19.1°$.
4	Determinant test: a_2-a_1=(1,1,1); b_1=(1,2,3); b_2=(2,3,4). det=1(2·4−3·3)−1(1·4−3·2)+1(1·3−2·2)=1(−1)−1(−2)+1(−1)=−1+2−1=0. ✓ Coplanar. Normal $n=b_1\times b_2$: $\hat{\imath}(8-9)-\hat{\jmath}(4-6)+\hat{k}(3-4)=-\hat{\imath}+2\hat{\jmath}-\hat{k}$. Plane through (1,2,3): −(x−1)+2(y−2)−(z−3)=0 → −x+2y−z=0 → x−2y+z=0. Verify (2,3,4): 2−6+4=0 ✓.
5	Line direction ratios (2,3,1) = normal to plane (since perpendicular to plane). Plane: 2(x−1)+3(y−2)+1(z−3)=0 → 2x+3y+z=11. Angle with xy-plane (normal=(0,0,1)): cos $\theta=\|n_1\cdot n_2\|/(\|n_1\|\|n_2\|)=\|1\|/\sqrt{14}=1/\sqrt{14}$. $\theta=\cos^{-1}(1/\sqrt{14})\approx74.5°$.

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